

Modelling and structure-tailored control of biogas plants fed on agro-food residues

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Abstract—The problem addressed in this paper is the efficient operation of anaerobic digesters for biogas production. We improve the generality of a literature model (the AMOCO) to account for two hydrolysis steps. We introduce an LFT-based parameter identification technique, and demonstrate the achievable match improvements by validating the model versus experimental data from bench scale reactors fed on corn and cattle manure. We finally propose an innovative scheme for methane production rate control, tailored to the observed dynamic structure of the process, and based on a cascade structure having VFA concentration as the inner variable. A simulation campaign on the model aligned to experimental data, shows that this scheme improves control quality over the state of the art, yet preserving simplicity and ease of tuning—a true necessity for field acceptance.

I. INTRODUCTION

The reckless use of fossil resources and the increase in waste production, have strongly fostered the quest for techniques to dispose of waste and produce energy. A successful one is the production of biogas from anaerobic digestion.

The complex biochemical processes in anaerobic digestion (hereinafter AD for short) require strict control of the operating conditions. In the absence of that, production may not just be “suboptimal”: it may even not start at all. In this respect, two major problems need facing: there are very few control-oriented models, and plant instrumentation is poor.

Concerning models, the most famous and successful one is the AD Model No. 1 (ADM1) [1]. It is very detailed, but of hardly any used for control design. Around one hundred parameters have to be identified, part of them depending on the substrate characteristics. The amount and quality of realistically available data is hardly ever sufficient.

As a consequence, simple models were proposed, focused on a subset of processes, or specialised to particular substrates. Among these, the most important is the AMOCO model [2], developed primarily as a tool to monitor and control the AD process, and achieving a good compromise between simplicity and precision.

As for control, the literature is almost silent, and there are very few applications in the field. Typical AD reactors are poorly instrumented, with few sensors, mostly of general-purpose industrial type. In general feeding is performed

discontinuously by the operator, based on empirical methods and sporadic laboratory analyses. The most used such method [9], is based on the FOS/TAC analysis, indicating the risk of inhibition by acidification. The analysis is done by titrating a sample manually taken from the digester, and requires 10 to 30 minutes. The results are used to determine whether and how much to feed.

In the context just sketched, the contributions of this paper are summarised below.

- We improve the flexibility of the AMOCO model, initially designed to describe the degradation of soluble organic matter and then modified to include a first-order hydrolysis phase [5], so as to account for *two* hydrolysis steps, relevant to two different particulate organic matters, namely corn and cattle manure sludge.
- We address parameter identification via Biochemical Methane Potential (BMP) tests applied to a selected set of parameters, the others being assumed based on ADM1. Most relevant, we show that a significantly improved match comes from applying LFT-based identification to an extended model parameter set.
- We validate the identified model versus data collected during a continuous experimental trial performed on bench-scale anaerobic digesters.
- We employ the model to tune and assess “classical” controls, based on single PI loops, and by using the model to also test off-design operating conditions, we evidence the limits of the said controls.
- Based on the so evidenced limits and the observed dynamic structure of the process, we propose a novel control scheme, based on a cascade configuration having the Volatile Fatty Acids (VFAs) concentration as inner variable, and we demonstrate its advantages.

The paper is organized as follows. Section II describes the modified AMOCO model. Section III illustrates the identification of the model based on BMP tests. Section IV describes the LFT-based parameter identification. Section V deals with control and presents the novel cascade-based structure. Section VI draws some conclusions, and sketches out future research.

II. PROCESS MODEL

Two hydrolysis dynamics are considered, relevant to the corn and cattle manure particulate components, X_{01} and X_{02} ; the dynamics of the non degradable matter components X_{nd} is also accounted for. The said components are expressed as volatile solids (VS) concentrations, as gVS L^{-1} .

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Like in the original AMOCO, only two bacterial populations are considered. In a first step, the acidogenic bacteria X_1 consume the soluble organic substrate and produce CO_2 and VFAs. In a second step, methanogenic bacteria use the VFAs as growth substrate and produce CO_2 and methane. The concentrations of acidogenic bacteria X_1 and methanogenic bacteria X_2 are also expressed as gVS L^{-1} , while COD (Chemical Oxygen Demand) is the unit for the concentrations of organic substrate S_1 and VFA S_2 , thus quantified as gCOD L^{-1} . The model inputs are the inlet flow rate q (L/s) and concentrations of corn silage, cattle manure particulates $\bar{X}_{01}$ and $\bar{X}_{02}$ (gVS L^{-1}), and cheese whey composed mainly by sugars readily biodegradable and soluble, hence expressed as substrate $\bar{S}_1$ (gCOD L^{-1}). The model has 7 differential equations: 2 for the hydrolysis steps, 1 for the dynamics of the fraction of non-degradable material, 2 for the mass balances of the bacterial populations, 2 for the organic substrate and the VFAs, i.e.,

$$\dot{X}_{01} = \frac{q}{V} (\bar{X}_{01} - X_{01}) - K_{h1} X_{01} \quad (1)$$

$$\dot{X}_{02} = \frac{q}{V} (\bar{X}_{02} - X_{02}) - K_{h2} X_{02} \quad (2)$$

$$\dot{X}_{nd} = -\frac{q}{V} X_{nd} + K_{h1} K_{nd1} X_{01} + K_{h2} K_{nd2} X_{02} \quad (3)$$

$$\dot{X}_1 = -\frac{q}{V} X_1 + \mu_1 \frac{S_1}{S_1 + K_{S1}} X_1 - K_{d1} X_1 \quad (4)$$

$$\dot{X}_2 = -\frac{q}{V} X_2 + \mu_2 \frac{S_2}{S_2 + K_{S2} + S_2^2/K_I} X_2 - K_{d2} X_2 \quad (5)$$

$$\dot{S}_1 = \frac{q}{V} (\bar{S}_1 - S_1) + K_{01} K_{h1} X_{01} + K_{02} K_{h2} X_{02} - K_1 \mu_1 \frac{S_1}{S_1 + K_{S1}} X_1 \quad (6)$$

$$\dot{S}_2 = -\frac{q}{V} S_2 + K_2 \mu_1 \frac{S_1}{S_1 + K_{S1}} X_1 - K_3 \mu_2 \frac{S_2}{S_2 + K_{S2} + S_2^2/K_I} X_2 \quad (7)$$

where V (L) is the volume, μ_i ($i = 1, 2$) (d^{-1}) are the maximum specific growth rates, K_{S1} , K_{S2} and K_{I2} (gCOD L^{-1}) are respectively the half-saturation and the inhibition constants, K_{hi} ($i = 1, 2$) (d^{-1}) are the hydrolysis kinetics, K_{di} ($i = 1, 2$) (d^{-1}) are decay constants, K_{ndi} ($i = 1, 2$) (gVS/gVS), K_{0i} ($i = 1, 2$) (gCOD/gVS) and K_i ($i = 1, 2, 3$) (gCOD/gVS) are stoichiometric coefficients. The methane production rate r_{CH_4} ($\text{mol}_{\text{CH}_4} \text{L}^{-1} \text{d}^{-1}$) is given by

$$r_{\text{CH}_4} = K_4 \mu_2 \frac{S_2}{S_2 + K_{S2} + S_2^2/K_I} X_2 \quad (8)$$

where K_4 ($\text{mol}_{\text{CH}_4}/\text{gVS}$) is a stoichiometric coefficient. Finally, the total concentration of volatile solids X_T is

$$X_T = X_{01} + X_{02} + X_{nd} + X_1 + X_2. \quad (9)$$

III. BMP-BASED PARAMETER IDENTIFICATION

A first model parameter estimation has been based on BMP tests, characterising the methane production for a specific substrate. In these tests, performed on the experimental setup described in the Appendix, small reactors are first filled with the substrate (cattle manure or corn) and with a

TABLE I
RESULTS OF BMP IDENTIFICATION

B_{01}	353 ($\text{NmL}_{\text{CH}_4}/\text{gVS}$)	K_{h1}	0.12 (d^{-1})
B_{02}	190 ($\text{NmL}_{\text{CH}_4}/\text{gVS}$)	K_{h2}	0.11 (d^{-1})
K_{01}	22.7 (gCOD/gVS)	K_{nd1}	0.23 (gVS/gVS)
K_{02}	12.2 (gCOD/gVS)	K_{nd2}	0.58 (gVS/gVS)

TABLE II
ADM1 PARAMETERS

μ_1	3.5 (d^{-1})	μ_2	0.74 (d^{-1})
K_{d1}	0.04 (d^{-1})	K_{d2}	0.02 (d^{-1})
K_{S1}	7.1 (gCOD/L)	K_{S2}	0.59 (gCOD/L)
K_1	28.6 (gCOD/gVS)	K_2	27.2 (gCOD/gVS)
K_3	30.3 (gCOD/gVS)	K_4	0.02 ($\text{mol}_{\text{CH}_4}/\text{gVS}$)

common inoculum of anaerobic bacteria taken from a full scale biogas plant fed on similar substrates; the methane production is then monitored, and its steady state value defines the potential of methane production of the substrate. The identification of the BMP curves characterises both the amount of methane produced, and the hydrolysis dynamics of the various types of substrates.

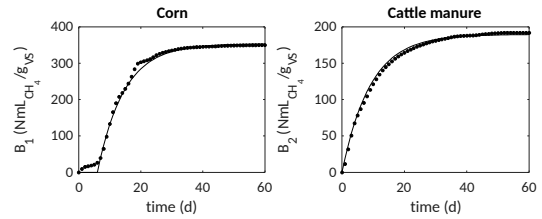


Fig. 1. Identification of BMP (data: dots, model: solid line)

Two tests were performed for each substrate, using the average trends for the identification of a first order dynamics ($i = 1$: corn, $i = 2$: cattle manure):

$$B_i(t) = B_{0i} (1 - \exp(-K_{hi}t)) \quad (10)$$

where $B_i(t)$ is the methane production ($\text{mol}_{\text{CH}_4}/\text{gVS}$), B_{0i} is the maximum methane production ($\text{mol}_{\text{CH}_4}/\text{gVS}$) and K_{hi} is the kinetic parameter (d^{-1}). The results are shown in Fig. 1 and Table I, where K_{0i} and K_{ndi} have been estimated from the values of B_{0i} and from the values of the ADM1 parameter (K_I has been actually set to avoid inhibition) as per Table II [6], [11].

The identified model has been validated with respect to experimental data collected on the two-reactor UIT-BTP2-Control setup (see again the Appendix) with continuous feeding and pH control. Methane production rate and composition were monitored online, while VFAs and volatile solids concentrations were measured manually. In Fig. 2 the simulated and experimental measurements of S_2 , r_{CH_4} and X_T , collected on reactor R1, are compared.

The VFAs production is largely underestimated by the model, while the concentration of volatile solids is overestimated. It is also important to note that the identified

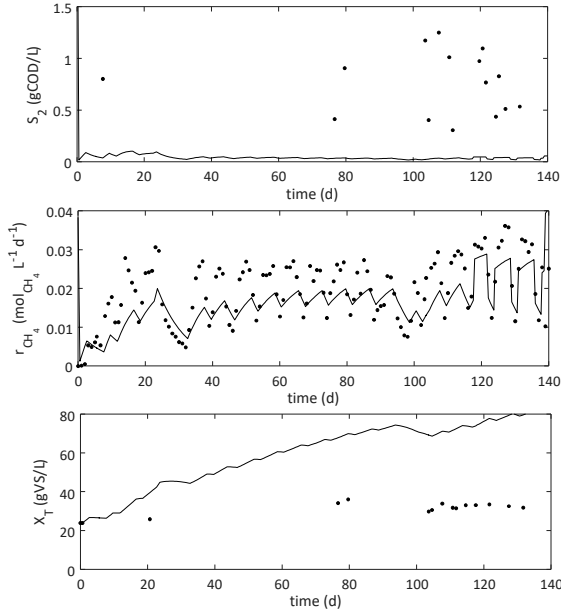


Fig. 2. Experimental and simulated measurements of S_2 , r_{CH_4} and X_T (data: dots, model: line).

model is characterized by a slower dynamics with respect to experimental data. A possible cause is the adaptation or “specialization” of the bacterial population to a constant and specific feeding. This may lead to an increase in production, especially in terms of speed, but also in the substrate degradability quantity.

IV. LFT-BASED PARAMETER IDENTIFICATION

To improve identification, a Linear Fractional Transformation (LFT) reformulation ([7]) of model (1–7) was adopted, see the Appendix. This allows a nonlinear, maximum likelihood, output-error estimate of parameters δ_i ([8]), where the output prediction and the error sensitivities to parameters δ_i can be computed without resorting to finite differences, by sampling the output of a time-variant, linear system ([5]).

TABLE III
RESULTS OF LFT IDENTIFICATION

B_{01}	389 (NmL $_{CH_4}$ /gVS)	K_{h1}	0.38 (d $^{-1}$)
B_{02}	215 (NmL $_{CH_4}$ /gVS)	K_{h2}	0.25 (d $^{-1}$)
K_{d1}	0.23 (d $^{-1}$)	K_{d2}	0.32 (d $^{-1}$)
K_{01}	24.9 (gCOD/gVS)	K_{nd1}	0.14 (gVS/gVS)
K_{02}	13.8 (gCOD/gVS)	K_{nd2}	0.56 (gVS/gVS)

A sensitivity analysis revealed that in the region of interest, K_{hi} and B_{0i} greatly influence the production of methane and less the production of VFAs and volatile solids, while decay parameters K_{di} affect, as expected, the amount of total solids and the production of VFAs. These considerations led to carry out two repeated identifications in cascade: the first based on the production of methane, for K_{hi} and B_{0i} only; the second based on all three outputs, for K_{di} . The identified parameters, based on the data collected on reactor R1, are reported in Table III; Fig. 3 shows the results of the

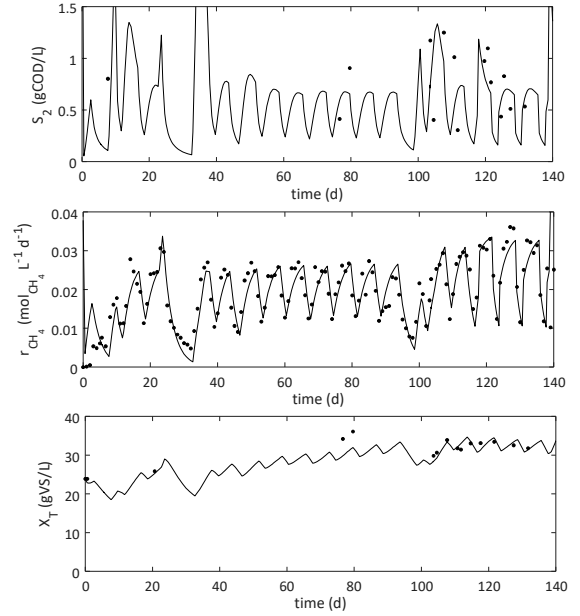


Fig. 3. Reactor R1: experimental and simulated measurements of S_2 , r_{CH_4} and X_T (data: dots, model: line)

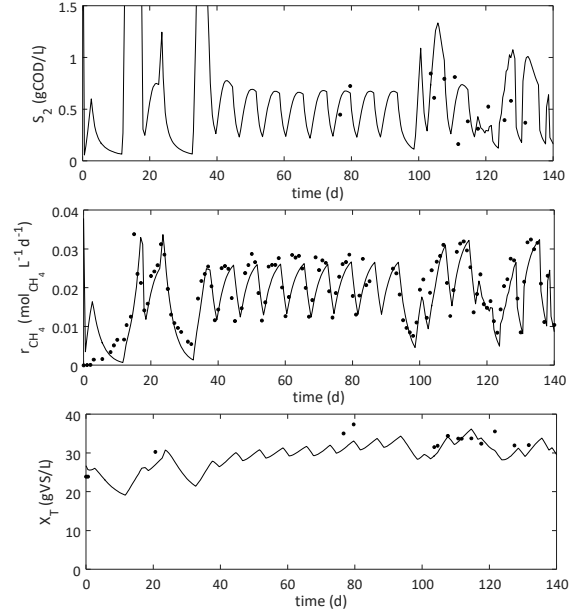


Fig. 4. Reactor R2: experimental and simulated measurements of S_2 , r_{CH_4} and X_T (data: dots, model: line)

identification, and Fig. 4 shows the validation with respect to the data collected on reactor R2.

The matching between simulations and experiments was also checked through TIC coefficients [13], computed for each output variable and reported in Table IV; a TIC value below 0.3 indicates a good match of the model with the measured data [13], [14].

V. CONTROL OF BIOGAS PRODUCTION

To regulate the biogas production, it is possible to add a whey flowrate w_3 (g/s) through the use of peristaltic pumps.

TABLE IV
TIC COEFFICIENTS

	Reactor R1	Reactor R2
S_2	0.21	0.25
r_{CH_4}	0.08	0.11
X_T	0.03	0.07

This allows to directly modulate the input variable $\bar{S}_1$:

$$q = \frac{w_1 + w_2 + w_3}{1000} \quad (11)$$

$$\bar{S}_1 = \frac{K_{03} K_{VS_3} w_3}{1000 q} \quad (12)$$

where w_1 and w_2 are the mass flowrates of corn and sewage respectively (considered as disturbances), $K_{03} = 22.1$ (gCOD/gVS) and $K_{VS_3} = 137$ (gVS/kg) is the whey volatile solids content.

A simple fixed parameters PI control with gain K and integral time T_i has been first tested by simulation ($e(t) = \bar{r}_{CH_4}(t) - r_{CH_4}(t)$). Parameters $K_P = 4.9 \times 10^4$ (gL/mol $_{CH_4}$) and $T_I = 4.5 \times 10^3$ (s) have been first determined by the IMC tuning method [10] and then finely manually tuned. A digital control has been implemented, with a sampling time of 5 minutes, and the results are reported in Fig. 5.

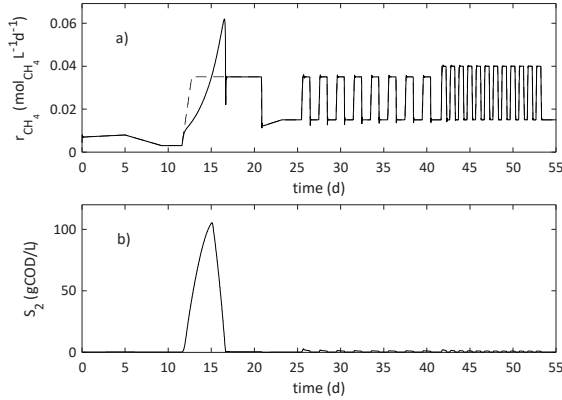


Fig. 5. a) Methane production rate (solid line) and set-point (dashed line), b) VFAs concentration

The sudden and high production rate demand at low loads causes a high feed flow rate, with consequent accumulation of VFAs (Fig. 5b), in turn responsible for the peak in methane production (Fig. 5a). However, the main problem is that high concentrations of VFAs may cause the inhibition of the process. A gain scheduling technique based on the actual value of methane flowrate has been also implemented, but no improvement was observed.

A significant improvement was conversely achieved by assuming that a measure of the VFA concentration S_2 is available. In this respect various techniques exist, including gas chromatography produced from an acidified sample [3], titration [4] and infrared spectrometry [12]). These can be implemented with appropriate instrumentation, more complex than a simple sensor, but economically feasible.

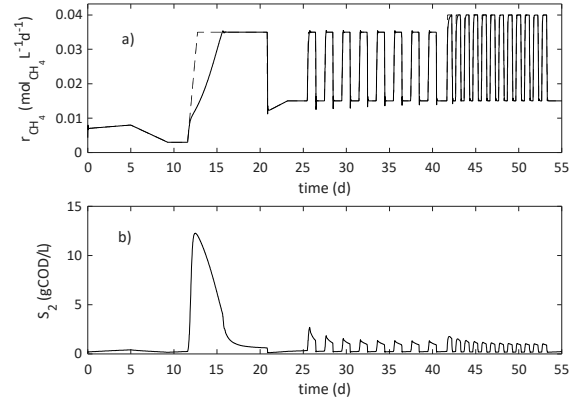


Fig. 6. a) Methane production rate (solid line) and set-point (dashed line), b) VFAs concentration

VFAs measurement was used first to implement a simple feedforward compensation in the control law:

$$w_3(t) = K_P e(t) + \frac{K_P}{T_I} \int_{t_0}^t e(\tau) d\tau + K_C S_2(t) \quad (13)$$

with $K_C = -1.7 \times 10^{-4}$ (gTQ L gCOD $^{-1}$ s $^{-1}$). In this way, for high values of S_2 , the negative gain in (13) reduces the control variable, thus limiting VFAs accumulation. The improvement is clearly appreciable in Fig. 6.b, where the peak in S_2 is reduced nearly by an order of magnitude.

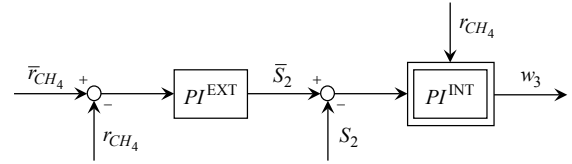


Fig. 7. Cascade control structure

The availability of a measure of VFAs and a more in-depth analysis of the model, however, have suggested a more complex cascade control structure (Fig. 7), capable of regulating both the production of methane and the concentration of VFAs. The goal is to control the acidogenesis and methanogenesis processes through the internal, variable-gain PI controller closed on the measurement of VFAs concentration, while the methane production rate is controlled by the external, fixed-gain PI.

The dynamics of the processes of acidogenesis, acetogenesis and methanogenesis depend not only on the concentration of S_2 and S_1 , but also on the concentrations of X_1 and X_2 which are not measurable. This entails relevant non-linearities. In particular, the response of the VFAs concentration to a step increase in the control variable is a temporary rapid accumulation, due to the fact that VFAs are produced by X_1 and consumed later by X_2 (i.e., they are intermediate products of a series of two processes). The amplitude of the peak depends on that of the step increase compared to the concentration of X_1 and X_2 in the reactors; however, the latter are not directly measurable, as stated before, but they are certainly dependent on the rate of biogas production: the

TABLE V
PARAMETERS OF EXTERNAL ($K_P^{\text{EXT}}, T_I^{\text{EXT}}$) AND INTERNAL (K_P', T_I^{INT})
PI REGULATORS

K_P^{EXT}	2.5×10^6 (gCOD s/mol $_{CH_4}$)	T_I^{EXT}	10^3 (s)
K_P'	2.5×10^4 (gTQ L (gCOD s) $^{-1}$)	T_I^{INT}	5×10^4 (s)

greater the methane production rate, the greater the bacterial population in the reactor and the lower the peak.

It was therefore decided to change the gain of the internal controller linearly with the methane flow rate, so that the controller has a more moderate response where the probability of accumulation of VFAs is greater, i.e.,

$$K_P^{\text{INT}}(r_{CH_4}) = K_P' r_{CH_4} \quad (14)$$

A further advantage deriving from this control strategy is the possibility of using a dynamic saturation on the external regulator, so as to limit the concentration of VFAs to a desired value, calculated in real-time based on empirical considerations and/or process inhibition conditions.

The Ziegler-Nichols rules were used to tune the parameters of the internal controller, while the IMC rules were used to tune the parameters of the external one. The obtained values were then slightly modified to better adapt to nonlinearities, see Table V.

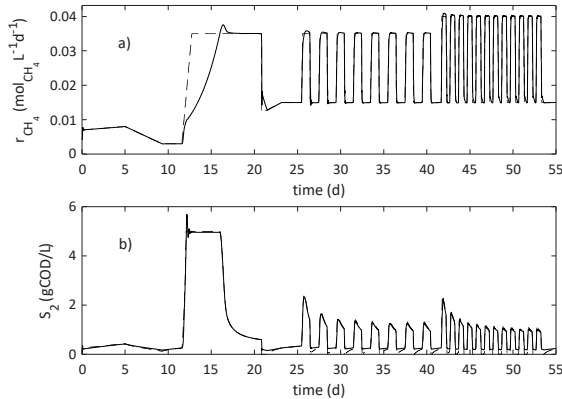


Fig. 8. a) Methane production rate (solid line) and set-point (dashed line), b) VFAs concentration (solid line) and set-point (dashed line)

The results obtained from cascade control are reported in Fig. 8. The further reduction of the VFAs peak is apparent.

Summarising, the model devised in the previous sections, validated on experimental data, allowed to figure out a control structure encompassing an industrially viable additional instrumentation, and yielding a significant increase in control quality while preserving an acceptable tuning simplicity. The technique was assessed in simulation; tests on the physical plant (lasting several months) are underway, and the results will be presented in future works.

VI. CONCLUSIONS AND FUTURE WORK

We extended the AMOCO model for AD to account for two hydrolysis steps, applying the extended model to a case with corn and cattle manure. The model was used to identify

the AD process starting from data obtained by batch and continuous experimentation on laboratory scale reactors.

We carried out the identification of parameters through two methods: the analysis of BMP, and an LFT approach. A better matching of the model with experimental data was obtained from the latter approach, which we therefore propose as the tool of election for this purpose.

Once the model was identified and validated, we used it to test methane production rate control strategies. Although the system is highly non-linear, we found that even simple PI controllers can provide an acceptable control quality. However, analysing their outcome, we figured out that the online measurement of VFAs concentration, employed as the internal variable of a cascade structure, is of great importance to improve the stability of digestion. We tested this novel strategy in simulation, which confirmed the assumption above.

Future work will be directed to testing the new strategy on a physical plant, and to extending the model and the experimentation to other AD cases.

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TABLE VI
UIT-BTP2 FEEDING SCENARIOS

	Scenario 1 0–97 07/16–10/21	Scenario 2 98–103 10/22–10/27	Scenario 3 104–137 10/28–11/30	Scenario 4 138–151 12/01–12/14	Scenario 5 152–159 12/15–12/22
OLR (gVS L ⁻¹ d ⁻¹)	1.42	1.7	1.7	2.01	2.45
Cattle manure (g/d)	150	97	110	110	134
Corn silage (g/d)	54	55	52	61	75
Cheese whey (g/d)	0	0	58	69	84
HRT (d)	58.8	28.8	58	50	41

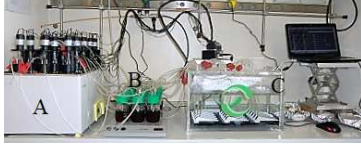


Fig. 9. AMTPS setup for batch BMP tests

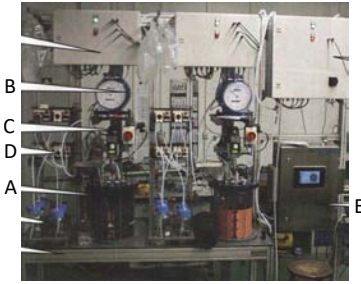


Fig. 10. UIT-BTP2 setup for continuous digestion tests

APPENDIX

Experimental setup for BMP tests. BMP tests were performed through an Automated Methane Potential Test System (AMTPS) by Biprocess Control (Fig. 9). Biogas production takes place in unit A, made by 15 0.5 L glass bottles kept at 35°, filled with sample, inoculum and amending solutions, under anaerobic conditions. Biogas is then purified (mainly from CO₂ and H₂S) in unit B and finally collected in unit C, where the CH₄ production is sampled.

Experimental setup for UIT-BT2 tests. The test plant UIT-BTP2 (Fig. 10) is made up by two 13.7 L reactors (A), equipped with a free access funnel for manual feeding, a manual drain valve, a pH probe, a temperature sensor, a connection for the output biogas, and a sampling port. The biogas is conveyed into a foam control chamber before going to the gas counter (B). The reactors are also equipped with stirrers (C) digitally controlled via PC. There are 4 peristaltic pumps for each reactor (D), for continuous/discontinuous feeding and discharge, for pH correction. After measuring the quantity, the gas is sampled in bags (E) to be analysed: CO₂ and CH₄ through infrared spectroscopy, O₂ and H₂S with electrochemical techniques.

The reactor was initially inoculated by digestate from an agricultural biogas plant. Reactors were fed with a mixture of corn silage and cattle manure. The organic loading rate

(ORL) and hydraulic retention time (HRT) was varied as per Table VI, feeding was done manually once per (working) day. The production and composition data of the biogas were obtained from the log system, while VS, soluble COD, VFA data per were measured on grab samples. In scenario N° 3 the supply differs for the two reactors, as for the second reactor (R2) a ramp profile in increments of 10%/day was applied to the OLR. Data collected during the first week of operation have always been reported but never used for identification purposes, the same for the data from the last scenario of reactor R2 due to a malfunction.

Formulation of the LFT model. Defining the vectors

$$x^T = [X_{01} \ X_{02} \ X_{nd} \ X_1 \ X_2 \ S_1 \ S_2] \quad (15)$$

$$y^T = [x_7 \ K_4\zeta_9 \ x_1 + x_2 + x_3 + x_4 + x_5] \quad (16)$$

$$u^T = [q \ \bar{X}_{01} \ \bar{X}_{02} \ \bar{S}_1] \quad (17)$$

$$\delta^T = [K_{h1} \ K_{h2} \ B_{01} \ B_{02} \ K_{d1} \ K_{d2}] \quad (18)$$

$$\rho^T = [2.21 \cdot 10^{-3} \ 2.05 \cdot 10^{-3} \ 64.2 \cdot 10^{-3}] \quad (19)$$

$$w^T = [\delta_1 z_1 \ \delta_2 z_2 \ \delta_3 z_3 \ \delta_4 z_4 \ \delta_5 z_5 \ \delta_6 z_6] \quad (20)$$

$$z^T = [x_1 \ x_2 \ w_1 \ w_2 \ x_4 \ x_5] \quad (21)$$

$$w^T = [u_1 \ u_2 \ u_3 \ u_4 \ x_1 \ x_2 \ x_3 \ x_4] \quad (22)$$

$$x_5 \ x_6 \ x_7] \quad (23)$$

$$\zeta = \begin{bmatrix} \omega_1(\omega_2 - \omega_5) \\ \omega_1(\omega_3 - \omega_6) \\ \omega_1\omega_7 \\ \omega_1\omega_8 \\ \omega_1\omega_9 \\ \omega_1(\omega_4 - \omega_{10}) \\ \omega_1\omega_{11} \\ \mu_1\omega_{10}\omega_8/(\omega_{10} + K_{S1}) \\ \mu_2\omega_{11}\omega_9/(\omega_{11} + K_{S2} + \omega_{11}^2/K_I) \end{bmatrix} \quad (24)$$

the model is written as

$$\dot{x} = \begin{bmatrix} \zeta_1/V - w_1 \\ \zeta_2/V - w_2 \\ -\zeta_3/V + w_1 - \rho_1 w_3 + w_2 - \rho_2 w_4 \\ -\zeta_4/V + \zeta_8 - w_5 \\ -\zeta_5/V + \zeta_9 - w_6 \\ -\zeta_6/V + \rho_3 w_3 + \rho_3 w_4 - K_1 \zeta_8 \\ -\zeta_7/V + K_2 \zeta_8 - K_3 \zeta_9 \end{bmatrix}. \quad (25)$$